Technical brief for expert review

# From column-wise HSE to 3D pressure relaxation

## Equations, implemented iteration, and a failed consistency test

Prepared 12 October 2026. This brief describes the current local Julia/C++ code and the measured 32 × 32 × 63 test. It requests human expert assessment; it does not claim a validated 3D equilibrium or present a literature review. References [S1]-[S7] identify source files and results.

Central finding
The 1D bootstrap is internally consistent with vertical hydrostatics, the production EOS and the optical-depth-to-height mapping. The best retained 3D iterate reduces the normalized worst-edge force residual, but violates vertical balance, EOS consistency and height-map consistency. The 3D state should not be treated as an acceptable equilibrium.

| Test configuration | Value |
|---|---|
| Input / geometry | Central crop of initial_atmosphere.h5; 63 optical-depth samples; 32 × 32 columns; dx = dy = 48 km. |
| Physics | Static nonmagnetic HSE; g = 274 m $s^{-2}$; prescribed temperature profiles on fixed log $\tau_{500}$. |
| Boundary and reference | $P_{top}$ = 0.1 Pa; $z_{top}$ = 0 in every column; no per-column shift at τ = 1. |
| Production closure | Wittmann EOS; native Julia continuum mass opacity at 500 nm. |
| Execution | Single rank, 8 Julia threads; 191 bootstrap iterations; 500 attempted 3D outer iterations. |

## Questions requiring expert judgment

Does the prescribed temperature-on-optical-depth atmosphere admit full nonmagnetic 3D HSE under these height and pressure anchors? If so, how should pressure, thermodynamics and geometry be solved together? Which residuals and consistency constraints must be enforced before accepting a reconstructed atmosphere?

The current code still returns accepted_3d = true and converged = false for this run, with status improved_unconverged_3d. “Accepted” means improvement in one numerical merit function, not physical validity. A strict rejection policy has been discussed but has not been implemented. [S1, S2, S6]

# 1. Governing equations and coordinates

Use SI units. P is gas pressure (Pa), ρ mass density (kg $m^{-3}$), $n_e$ electron number density ($m^{-3}$), $\kappa_{500}$ mass extinction ($m^2$ $kg^{-1}$). z increases upward. Gravity is -g $e_z$. No velocity, inertia or energy equation enters this reconstruction.

$$\nabla P = -\rho g\, e_z \; ; \; P_x|_z = 0 \; , \; P_y|_z = 0 \; , \; P_z = -\rho g \tag{1}$$

The MHS branch, not used in this test, replaces the right side by $q = F_L - \rho g\, e_z$, with $F_L = [(\nabla \times B) \times B]/\mu_0$. The supplied B samples are held fixed in optical depth; the reconstruction does not solve for B or enforce $\nabla \cdot B = 0$. HE3D is selected only when the magnetic field is absent. [S1:1-6, 69-108]

$$(\rho, n_e) = \mathrm{EOS}(T, P) \; ; \; \kappa_{500} = \kappa(T, P) \; ; \; d\tau_{500}/dz = -\kappa_{500}\rho \tag{2}$$

EOS(T,P) is the production Wittmann implementation, not the ideal-gas fixture used in some tests. The C++ adapter obtains electron pressure from gas pressure, derives density and $n_e$, and converts CGS quantities to SI. The opacity routine independently reconstructs a continuum state from T and P, then divides extinction per metre by EOS density to obtain mass opacity. [S3, S4]

## The stored grid is corrugated in physical space

Array coordinates are (s, x, y), where $s = \log_{10} \tau_{500}$. The s grid is fixed and z = z(s,x,y) is reconstructed. A horizontal index edge generally has $\Delta z \neq 0$. Temperature is fixed at each optical-depth sample, not at a fixed geometrical height.

$$a_z = a_s/z_s \; ; \; a_x|_z = a_x|_s - z_x|_s\, a_s/z_s$$
$$a_y|_z = a_y|_s - z_y|_s\, a_s/z_s \tag{3}$$

The code uses these chain-rule derivatives for magnetic curl calculations, with centered two-neighbor differences inside and one-sided differences at endpoints. Pressure balance itself is evaluated by physical edge integrals. [S1:69-108]

Geometrical anchors
The height integration sets the smallest-optical-depth plane to z = 0 in every column. There is no freely solved column height offset. z(log τ = 0) is not forced to zero. Thus a change of column thickness cannot be explained by a different τ = 1 zero-point convention.

# 2. Column-wise 1D HSE bootstrap

Each column first solves vertical HSE, coupled iteratively to EOS and opacity, without magnetic force or horizontal pressure constraints. The initial run has P = 0.1 Pa and ρ = $10^{-10}$ kg $m^{-3}$ throughout the volume; the density value is an initialization, not an enforced density boundary. [S1:606-641; S2:293-351]

$$z^*_{k+1} = z^*_k - \Delta\tau_k / [ (\kappa^*_k\rho^*_k + \kappa^*_{k+1}\rho^*_{k+1})/2 ] \quad (4)$$

Here k progresses from top to bottom, Δτ > 0 and $z^*_{top} = 0$. This is Δτ divided by the endpoint-averaged extinction, not a trapezoidal average of inverse extinction. [S1:116-127]

$$P^*_{k+1} = P^*_k + g(\rho^*_k + \rho^*_{k+1})/2 \times |z^*_{k+1} - z^*_k| \quad (5)$$

$P^*_{top}$ is fixed to 0.1 Pa. Integration raises an error if a pressure becomes non-positive. The current experiment uses only a top pressure boundary. [S1:130-142]

## Exact update sequence

```
for each bootstrap iteration n:
    rho*, ne* = EOS(T, P[n])
    kappa*    = opacity500(T, P[n])
    z*        = height_from_tau(kappa*, rho*)
    P*        = vertical_hydrostatic_integral(rho*, z*)
    measure changes against the previous state
    P[n+1]   = 0.7 P*   + 0.3 P[n]
    rho[n+1] = 0.7 rho* + 0.3 rho[n]
    ne[n+1]  = ne*
    z[n+1]   = z*
```

## Bootstrap stopping and checkpoint

The stopping test requires maximum relative pressure and density changes ≤ $10^{-6}$ and maximum absolute height change ≤ $10^{-3}$ m. It does not require horizontal force balance. The maximum budget is 1,000 iterations. A finite, positive, monotonic, non-divergent state may also be returned at that limit, with bootstrap_converged = false.

For this run the bootstrap converged after 191 iterations. Its raw final pressure change was 3.79 × $10^{-8}$, density change 4.30 × $10^{-8}$, and height change 9.86 × $10^{-4}$ m. A snapshot is saved before any 3D relaxation and has independent copies of P, ρ, $n_e$ and z. [S6, S7]

# 3. The active 3D pressure relaxation

Define $r_v = (x_i, y_j, z_{kij})$ and $q = F_L - \rho g\, e_z$. For each neighboring edge $a \to b$, the target constraint is the trapezoidal physical line integral [S1:144-228; S2:118-187]:

$$P_b - P_a = I_{ab} \; ; \; I_{ab} = (q_a + q_b)/2 \cdot (r_b - r_a) \tag{6}$$

$$I^x_{ab} = F_x \, \Delta x + (F_z - g\bar{\rho}) \, \Delta z$$
For HSE: $\Delta P = -g\bar{\rho} \, \Delta z$ along a horizontal index edge. (7)

This is not an instruction to make neighboring optical-depth samples have equal pressure: their physical heights may differ.

## Jacobi-like neighbor sweeps

$$P_v^{s+1} = \max[\, \varepsilon_P, \Sigma_{u \in N(v)} w_{uv} (P_u^{\,s} + I_{uv}) \,/\, \Sigma_{u \in N(v)} w_{uv} \,] \tag{8}$$

The sweep reads the old buffer and writes a new buffer. Vertical weights are 1; x/y weights are $\lambda_n = \min(1, n/10)$ during continuation, then 1. The active relaxation does not divide these weights by physical edge length. Pressure is clipped to the numeric Float64 epsilon, about $2.22 \times 10^{-16}$ Pa. The top plane is reset to the prescribed pressure.

The code performs at least 10 and at most 200 sweeps. Inner convergence tests $\max |P^{s+1} - P^s| / \max(|P^s|, 1\ \text{Pa}) \le 10^{-6}$. The distributed path checks this every 10 sweeps and at the cap. Side and bottom options are named force_neumann; implementation omits exterior neighbors, with no periodic wrapping and no separately prescribed side/bottom pressure maps.

## Outer coupling and backtracking

```
rho*, ne* = EOS(T, P[n]);  kappa* = opacity(T, P[n])
z* = height_from_tau(kappa*, rho*)
P* = vertical_seed(rho*, z*)
P* = neighbor_pressure_relaxation(P*, rho*, z*, lambda[n])
for alpha = 0.7, 0.35, ... subject to configured limits:
    P_trial   = alpha P*   + (1-alpha) P[n]
    rho_trial = alpha rho* + (1-alpha) rho[n]
    ne_trial  = alpha ne*  + (1-alpha) ne[n]
    z_trial   = alpha z*   + (1-alpha) z[n]
    evaluate full edge force residual on trial state
```

Crucial ordering: $\rho^*$, $n_e^*$ and $z^*$ derive from P[n], while $P^*$ has already been changed by the pressure solve. They are then blended independently. EOS and geometry are not re-enforced at the trial pressure. [S1:727-766; S2:424-485]

# 4. Residuals, stopping and “acceptance”

## Force residual: a global maximum over physical edges

$$R_e = |\Delta P_e - I_e| / (\ell_e S) \; ; \; R = \max_e R_e$$
$$S = \max[g \max_{cells}|\rho|, \max_{cells}|F_L|] + \varepsilon \quad (9)$$

$\ell_e$ is physical edge length, including corrugation. The x/y/z diagnostic values group index edges; they are not Cartesian force components measured at fixed z. The whole-domain scale changes as density changes. A lower normalized R does not by itself prove all physical errors decreased. [S1:279-308, 642-673; S2:189-270]

## Compatibility and EOS diagnostics

$$C_f = |\Sigma_{e\in\partial f} \text{signed } I_e| / (\Sigma_{e\in\partial f}|I_e| + \varepsilon)$$
$$C_{max} = \max C_f \; ; \; C_{rms} = \sqrt{[\Sigma C_f^2/N_{faces}]} \quad (10)$$

Nonzero closed-face circulation demonstrates discrete incompatibility with an exact scalar pressure gradient. Zero circulation is necessary, not sufficient, for the coupled equilibrium. Near-zero edge forces can yield large normalized circulation ratios. Each of 183,551 faces is counted once across rank boundaries. [S1:230-276; S2:189-223]

$$E_\rho = \max |\rho_{stored} - EOS_\rho(T,P)| / \max(|EOS_\rho(T,P)|, \varepsilon)$$
$$E_{ne} \text{ is defined identically for electron density.} \quad (11)$$

These are relative ratios: 3.67 corresponds to 367% discrepancy. The added EOS and circulation checks report the state; they do not enter acceptance or convergence decisions. [S1:311-337]

## Three different decisions in the code

| Decision | Actual requirement |
|---|---|
| Accept a trial step | R_trial ≤ max(10^-5, 1.25 R_current). Up to 25% residual growth is therefore permitted; alpha is halved on failure. |
| Declare 3D convergence | λ = 1; relative P and ρ changes ≤ 10^-6; R ≤ 10^-5; max height change ≤ 10^-3 m. No explicit EOS or height-map consistency test. |
| Return a best 3D state | Keep an iterate with the lowest R. accepted_3d is true if it improves the bootstrap R by more than the machine-epsilon criterion, even when converged is false. |

Maximum outer budget: 500. If no improving iterate exists, the code restores the bootstrap. Otherwise it returns the best improvement, including an unconverged one. This run returns iteration 86 as the best state after attempting all 500 iterations. The final attempted iterate had R ≈ 0.954. [S1:755-807; S2:471-559; S6]

# 5. Measured bootstrap and 3D results

Same fixed temperature samples, coordinates, gravity, boundary pressure and linear update. The rerun exactly reproduced the earlier pressure, density, electron density and height arrays. No log-pressure or global-pressure experiment is active. [S6, S7]

| Metric | Bootstrap | Best returned 3D |
|---|---|---|
| Reported convergence | Yes (1D change test) | No |
| Iterations | 191 | 500 attempted; best 86 |
| Relative P / ρ step change | 3.79 × 10^-8 / 4.30 × 10^-8 | 3.78858 / 0.139879 |
| Max height step change | 0.000986 m | 374,550 m |
| Full / x residual | 0.981292 | 0.471630 |
| y residual | 0.960158 | 0.422020 |
| Vertical residual | 4.91 × 10^-12 | 0.468398 |
| Circulation max / RMS | 0.987913 / 0.060790 | 0.610316 / 0.036401 |
| Density EOS error | 3.97 × 10^-8 | 3.67447 (367%) |
| Electron EOS error | 2.15 × 10^-8 | 2.30993 (231%) |

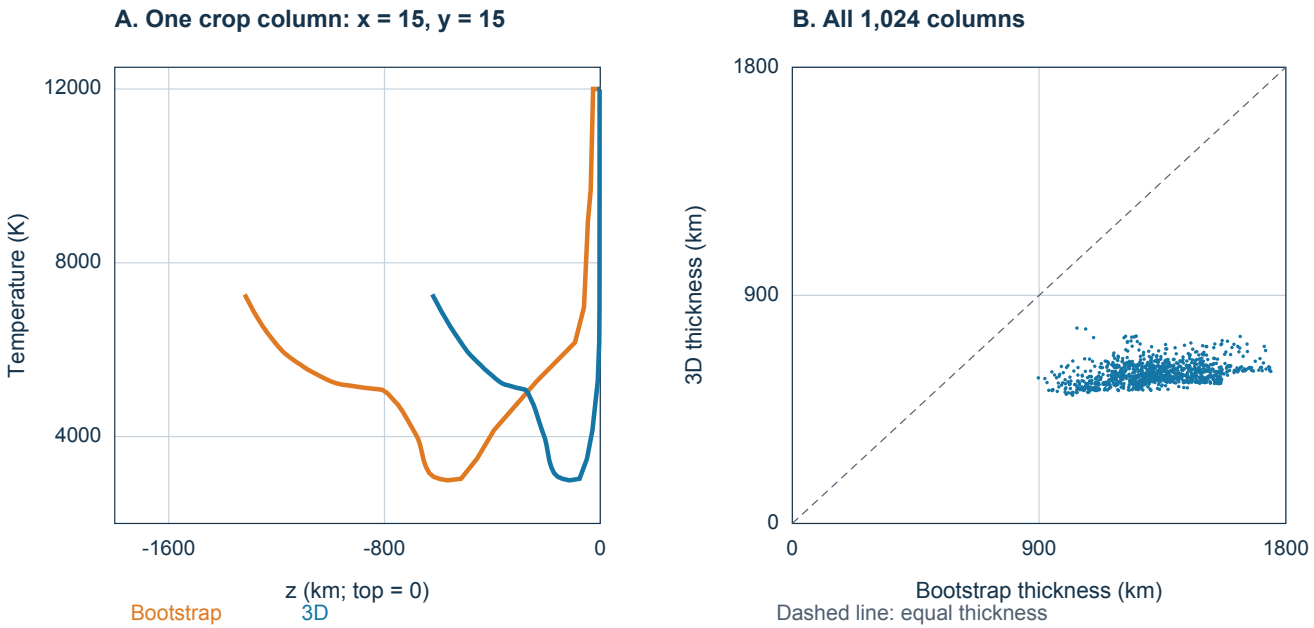



Figure 1. Recreated directly from saved HDF5 arrays. Panel A uses Python crop indices x = y = 15 (f[:,15,15]); it is an illustrative column, not an asserted match to the user's original plotted column. T(τ) is identical in both states. Panel B compares top-to-bottom thickness for every column. z is in metres in the files and kilometres only in this figure.

Bootstrap column thicknesses span 897-1,745 km; the returned 3D thicknesses span 506-771 km. For the illustrated column the thickness drops from 1,318 km to 623 km. There is no coded 620 km ceiling. The saved temperatures are unchanged; the altered geometric mapping compresses the same thermal profile.

# 6. Why this 3D state is not acceptable

## Independent reconstruction of the saved height mapping

An offline consistency test recomputed EOS(T,P) and κ(T,P) from each saved atmosphere, then reintegrated the same fixed τ grid with the same z_top = 0 and the same discrete formula (4). This is a consistency test, not a new force-balance solve. [S7]

$$\hat{z} = H[\,\tau,\ \kappa(T,P),\ EOS_{\rho}(T,P),\ z_{top} = 0\,]$$
$$E_{z,map} = \max_{cells} |z_{saved} - \hat{z}| \qquad (12)$$

| Measurement | Bootstrap | Returned 3D |
|---|---|---|
| Maximum mapping discrepancy | 0.000908 m (0.91 mm) | 613,415 m (613.4 km) |
| Worst 3D discrepancy location | - | Python f[62,25,20] |
| At that 3D location | - | z_saved = -599.1 km;<br>$\hat{z}$ = -1,212.6 km |

The largest 3D mismatch is at the bottom of local crop column x = 20, y = 25 (source x = 256, y = 261 in zero-based coordinates). Bootstrap and final files use the same top reference, so this difference is not a height-origin offset.

## Code-observed coupling issue

Independent blending generally does not preserve ρ = EOS(T,P), n_e = EOS_ne(T,P), or z = H(T,P). The density and height candidate lag the pressure candidate by one outer iteration. At convergence that lag should disappear; at the retained, unconverged iterate it has not. This accounts for a direct consistency failure without establishing that it is the only cause of poor convergence.

Increased extinction κρ reduces |Δz| for fixed Δτ, providing a mechanism for column compression. However, the saved z also fails reconstruction from its own T and P. It is therefore not justified to interpret the compressed profile as a converged physical atmosphere.

## Validity versus numerical merit

Observed, not hypothetical
The 3D state has a vertical residual of 0.468, large EOS discrepancies, a 613 km mapping inconsistency and converged = false. Finite positive pressure/densities and monotonic heights are necessary checks but do not make this state a valid HSE solution. Our current assessment is to reject it as an equilibrium; the software still labels it an accepted improvement.

The bootstrap is valid only as a numerical column-wise HSE reconstruction with these inputs and boundary assumptions. Its accuracy here is not evidence that all heights are independently established observationally, nor that it satisfies full 3D hydrostatics.

# 7. Questions for the expert

## A. Does a compatible full 3D HSE state exist?

$$\text{For constant } g\text{: } \nabla \times (-\rho g\, e_z) = 0 \Rightarrow \partial\rho/\partial x|_z = \partial\rho/\partial y|_z = 0 \tag{13}$$

This necessary continuum condition follows from $\nabla \times \nabla P = 0$. Together with $P = P(z)$, it imposes strong horizontal constraints on EOS(T,P). Given fixed $T(\tau,x,y)$, common top z and common top pressure, can a mapping $z(\tau,x,y)$ satisfy these constraints? Is the problem over-constrained for arbitrary input profiles, or are additional physical forces or degrees of freedom required? This is a mathematical concern to assess, not a proof of nonexistence for the tested atmosphere.

## B. How should thermodynamics and geometry be coupled?

Should P be the independent variable and ρ, n_e, κ and z be recomputed from the same trial pressure before evaluating its merit? Would that preserve the required mapping but destabilize the pressure iteration? Would a joint nonlinear solve or a different primary variable be more appropriate?

## C. What geometry and boundary freedoms are justified?

Should relative column height offsets be inferred rather than imposing a flat top optical-depth surface? If offsets are introduced, what observational or boundary information constrains them? How should lateral and bottom conditions be formulated on this corrugated grid?

## D. Which discrete operator and merit function are appropriate?

Are trapezoidal edge forces and the unweighted neighbor pressure estimates compatible with the intended continuous equation on a highly nonuniform optical-depth grid? Should weights account for edge lengths or cell volumes? What role should face circulation and dimensional residuals play alongside a globally normalized maximum?

## E. What constitutes an acceptable returned atmosphere?

Please propose joint acceptance and stopping criteria for full force balance, EOS consistency, optical-depth mapping consistency, positivity and monotonic geometry. Should the solver return an explicit failure rather than an unconverged “best improvement”? Can a physically valid 1D fallback be clearly distinguished from a successful 3D reconstruction?

Two preliminary trials, already removed
Pressure-only logarithmic blending stopped after two attempted outer iterations: best R = 0.855, with EOS errors ≈ 10^12. Existing global weighted pressure projection gave best R = 0.700 after 500 iterations; every step used a positive-pressure fallback, and none of those constrained inner solves converged within 1,000 iterations. These results assess those implementations/configurations, not the general validity of log variables or global methods.

# 8. Reproducibility and source map

Prepared from the current local working tree, including diagnostic/checkpoint additions. Repository HEAD: d92d724dcd4fbd6ecbb68e27168978c53aed780d. The working tree is modified; this hash alone does not reproduce the reviewed source. SHA-256 fingerprints below identify the exact snapshots inspected. This PDF contains analysis and figures, not the full datasets or source code.

## Code references (relative to repository root)

| ID | File and relevant implementation |
|---|---|
| S1 | FFNOInversion.jl/src/ForceBalance.jl<br>Options 26-42; coordinates/Lorentz 69-108; height/vertical pressure 116-142; pressure relaxation 144-212; diagnostics 230-337; bootstrap 606-641; outer reconstruction 674 onward. |
| S2 | FFNOInversion.jl/src/DistributedForward.jl<br>Pressure relaxation 118-187; halo-backed diagnostics 189 onward; bootstrap 293 onward; reconstruction 353 onward. This run uses that path with a serial context. |
| S3 | FFNOInversion.jl/src/EOS.jl:21-32<br>scripts/witt_eos_cpp.cpp:129-144, 644 onward<br>Production EOS and CGS/SI adapter. |
| S4 | FFNOInversion.jl/src/Opacity500.jl:8-15<br>Native continuum opacity uses continuum_state and continuum_extinction_m. |
| S5 | outputs/hse3d_bootstrap_relaxation32_20261012/run_hse3d.jl<br>Crop, units, options, checkpoint saving, output checks. |

## Data and measured evidence

Run directory: outputs/hse3d_bootstrap_relaxation32_20261012/
S6: diagnostics.toml, diagnostics.txt, iterations.log, stage_comparison.txt.
S7: atmosphere_bootstrap.h5 and atmosphere_hse3d.h5; verify_and_compare.jl; offline height-map test exported to tmp/pdfs/hse_expert/evidence.toml. Bootstrap output was regenerated separately and matched every prior bootstrap diagnostic exactly. Final arrays exactly reproduced the earlier step-1 run.

Original input: /Users/harshm/Documents/WorkRepo/ffnoml_forward_001/inputs/initial_atmosphere.h5. Source shape in Python is (63,504,504). Crop x/y are inclusive zero-based 236:267; all depths retained. log τ_500 spans approximately -7.8 to 1.0. Temperature spans 2,499.86-12,000 K. Output arrays in Python are (z,y,x) = (63,32,32); in Julia they are (z,x,y).

## Snapshot fingerprints (SHA-256; first 24 hexadecimal characters)

| Snapshot | Fingerprint prefix |
|---|---|
| ForceBalance.jl | d8cd1a671864078814133c31 |
| DistributedForward.jl | 610141ff8ab30ce21715185a |
| EOS.jl | 6f37b313c788dd0bc6a0ee00 |
| Opacity500.jl | a27d43cac3c6e16c6d523dd3 |
| atmosphere_bootstrap.h5 | 1fdd3b6482d26c87a2acf0da |
| atmosphere_hse3d.h5 | 856a54be9c90433fc24f1590 |

For expert review, supply the two HDF5 files, the listed implementation files, the EOS backend source/partition-function input, and the run script alongside this PDF. The supplied run uses one rank and 8 threads. Manufactured diagnostics/checkpoint tests and one-/four-rank parity checks pass; they do not certify this unconverged 3D atmosphere as an equilibrium.